Midterm 1 Review Guide
STAT 132; Winter 2025

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1 Basics

- Population: A **population** is a full set of units that we are interested in studying.
- Sample: A **sample** is a subset of the units in a population, for which we have measurements or data.
- Parameter: A **parameter** is numerical measurement that describes some characteristic of a population.
- Frequentist Inference: In the **frequentist** approach to statistics, parameters are considered fixed but unknown values.
- Bayesian Inference: In the **Bayesian** framework, parameters are treated as random variables that have their own probability distributions. These distributions reflect your uncertainty about the parameter before seeing any data (the "prior") and are updated as data becomes available (yielding the "posterior").
- Likelihood Function:

$$L(\theta|\boldsymbol{X}) = \prod_{i=1}^{n} f(X_i|\theta)$$

2 Bayesian Inference

- Prior: Suppose that you have a statistical model with parameter θ. If you treat θ as random, then the distribution that you assign to θ before observing the data is called its **prior distribution**.
- Posterior: Consider a statistical inference problem with parameter θ and random variables $X_1, \ldots, X_n$ to be observed. The conditional distribution of θ given $X_1, \ldots, X_n$ is called the **posterior distribution** of θ .
- Bayes Theorem: Suppose that the n random variables $X_1, \ldots, X_n$ form a random sample from a distribution for which the pdf or pmf is $f(x|\theta)$. Suppose also that the value of the parameter θ is unknown and the prior pdf or pmf of θ is $\pi(\theta)$, with parameter space $\theta \in \Theta$. Then the posterior pdf or pmf of θ is

$$\pi(\theta|x_1, \ldots, x_n) = \frac{f(x_1|\theta) \cdots f(x_n|\theta)\pi(\theta)}{m_n(x_1, \ldots, x_n)}$$

where $m_n(x_1, \ldots, x_n)$ is the joint marginal distribution of the data, $x_1, \ldots, x_n$.

- Conjugacy: For a given likelihood, if the prior and posterior distribution belong to the same family, this is said to be a **conjugate** prior distribution. In this case, posterior updating can be done simply by updating the parameters that correspond to the family.

- Beta-Bernoulli Conjugacy: Suppose that $X_1, \ldots, X_n$ form a random sample from the Bernoulli distribution with parameter p , which is unknown ($0 < p < 1$). Suppose also that the prior distribution of p is the beta distribution with parameters $\alpha > 0$ and $\beta > 0$. Then the posterior distribution of p given $x_1, \ldots, x_n$ is the beta distribution with parameters $\alpha + \sum_{i=1}^{n} x_i$ and $\beta + n - \sum_{i=1}^{n} x_i$.

- Gamma-Poisson Conjugacy: Suppose that $X_1, \ldots, X_n$ form a random sample from the Poisson distribution with parameter λ , which is unknown ($\lambda > 0$). Suppose also that the prior distribution of is the gamma distribution with parameters $\alpha > 0$ and $\beta > 0$. Then the posterior distribution of λ given $x_1, \ldots, x_n$ is the gamma distribution with parameters $\alpha + \sum_{i=1}^{n} x_i$ and $\beta + n$.

- Normal-Normal Conjugacy: Suppose that $X_1, \ldots, X_n$ form a random sample from a normal distribution for which the value of the mean μ is unknown and the value of the variance $\sigma^2 > 0$ is known. Suppose also that the prior distribution for μ is the normal distribution with mean μ_0 and variance σ_0^2. Then the posterior distribution of μ given $x_1, \ldots, x_n$ is the normal distribution with mean μ_1 and variance σ_1^2 where

$$\mu_1 = \frac{\sigma^2 \mu_0 + n\sigma_0^2 \bar{x}_n}{\sigma^2 + n\sigma_0^2}$$

and

$$\sigma_1^2 = \frac{\sigma^2 \sigma_0^2}{\sigma^2 + n\sigma_0^2}.$$

- Loss Function: A loss function is a real-valued function of two variables, $L(\theta, a)$, where $\theta \in \Theta$ and a is a real number. The interpretation is that the statistician "loses" by $L(\theta, a)$ if the parameter equals θ and the estimate equals a.

- Posterior Expected Loss: $\mathbb{E}\left[L(\theta, a)|\boldsymbol{x}\right] = \int_{\Theta} L(\theta, a)\pi(\theta|\boldsymbol{x})d\theta$.

- Bayes Estimator: Let $L(\theta, a)$ be a loss function. For each possible value x, let $\delta^*(\boldsymbol{x}) = a$ be such that $\mathbb{E}\left[L(\theta, a)|\boldsymbol{x}\right]$ is minimized. Then $\delta^*(\boldsymbol{x})$ is called a **Bayes estimator** of θ. Once $\boldsymbol{X} = \boldsymbol{x}$ is observed, $\delta^*(\boldsymbol{x})$ is called a **Bayes estimate** of θ.

- Bayes Estimator for Squared Error Loss: Let θ be a real-valued parameter. Suppose that the squared error loss function is used and that the posterior mean of θ, $\mathbb{E}(\theta|\boldsymbol{x})$, is finite. Then, the Bayes estimator of θ is $\delta(\boldsymbol{X}) = \mathbb{E}(\theta|\boldsymbol{X})$.

- Bayes Estimator for Absolute Error Loss: Let θ be a real-valued parameter. Suppose that the absolute error loss function is used. Then, the Bayes estimator of θ is the posterior median.

3 Maximum Likelihood Estimation

- MLE: The **maximum likelihood estimator** is defined as

$$\hat{\theta}(\boldsymbol{X}) = \arg_{\theta} \max L(\theta|\boldsymbol{X}),$$

provided it exists. After $\boldsymbol{X} = \boldsymbol{x}$ is observed, $\hat{\theta}(\boldsymbol{x})$ is called the **maximum likelihood estimate** of θ.

- General Procedure for MLE:
  1. Compute the log likelihood function: $\ell(\theta|\boldsymbol{x}) = \log\left[L(\theta|\boldsymbol{X})\right]$
  2. Differentiate the log likelihood with respect to θ and set this equal to zero: $\frac{d}{d\theta}\ell(\theta|\boldsymbol{x}) = 0$
  3. Solve this expression for θ - this is your candidate MLE ($\hat{\theta}$)
  4. Check that $\frac{d^2}{d\theta^2}\ell(\theta|\boldsymbol{x})|_{\hat{\theta}} < 0$ to ensure that this is a maximum and not a minimum
  5. Consider edge cases (i.e., does the MLE fall on the boundary of the parameter support)
- Invariance: Let $\hat{\theta}$ be a maximum likelihood estimate of some parameter θ. Also let $g(\theta)$ be any function of the parameter θ. Then the maximum likelihood estimate of $g(\theta)$ is $g(\hat{\theta})$.
- Consistency: Let $\hat{\theta}_n$ be the maximum likelihood estimate for some parameter θ using a sample size of n. Also, let θ_T represent the true unknown parameter value. Then, as $n \to \infty$, we have that $\hat{\theta}_n \xrightarrow{P} \theta_T$.